



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	B.TECH	IIT Kharagpur	9.41 / 10
2023	ISC Class XII Exam	Vivekananda Mission School	98.25%
2021	ICSE Class X Exam	Vivekananda Mission School	98.4%

INTERNSHIPS AND PROJECTS

Research Intern | IIT Kharagpur | Prof. Abhishek Sharma [Apr'25-May'25]
• Modelled and optimized a **closed-loop, dual-product** supply chain network with **3 CDRCs** and **10** customer zones
• Built a bi-objective model using **NSGA-II (68 gens)** and **MILP (1000 epsilon-runs)** to minimize cost and emissions
• Achieved **INR 327.10 million** and **36.92M kg CO₂** with **2.1%** deviation across solvers, validating solution consistency
• Visualized **Pareto front** evolution, extracted customer CDRC mappings, and generated computational reports

Last-Mile ETA Prediction & Route Optimization | Self Project [May'25-Jun'25]
• Generated **10,000** realistic delivery records with noise; clustered zones using **K-Means** and mapped them via **Folium**
• Reduced **MAE** by **50%** and improved **R²** to **0.53** by tuning **Linear, Ridge, RF, and XGBoost** models via **GridSearchCV**
• Improved ETA accuracy by **71%** & **R²** to **0.83** by implementing probabilistic models - **LightGBM Quantile & NGBoost**
• Simulated risk-aware routes using **OR-Tools**; visualized **5 & 6** stop plans on **Folium** maps and deployed via **Streamlit**

Quantitative Stock Forecasting | Self Project [Nov'24-Dec'24]
• Developed a stock forecasting model using **GBM, Heston, and 300 Monte Carlo runs** on real and synthetic data
• Verified assumptions via **D'Agostino-Pearson & ARCH** tests, confirming **heteroskedasticity** in real world stock data
• Compared outputs with **1.5% mean difference**, observing Heston's **10% tighter confidence bands** under real volatility
• Evaluated forecast quality via **Sharpe Ratio**, distribution plots, and variance diagnostics to assess model risks

WhatsApp Chat Analyzer | Self Project [Jul'24-Aug'24]
• Automated a Python tool to extract insights from raw WhatsApp chats using **regex, Pandas, and EDA techniques**
• Extracted insights on user activity, message frequency, and text patterns using **Matplotlib, Seaborn, and wordclouds**
• Deployed an interactive **Streamlit app** enabling dynamic analysis of group/private chats with **user-level filtering**

COMPETITION/CONFERENCE

Behavioural Credit Risk Modelling | Convolve 3.0 | IITB Case Study [Jan'25]
• Trained a behaviour scoring model on **96,806** credit accounts to predict default risk for **41,792 unseen** customers
• Engineered data pipeline with **median imputation, correlation filtering, and class reweighting** to address imbalance
• Benchmarked **CatBoost** and **XGBoost**, with **CatBoost** achieving **ROC-AUC of 87.2%** vs. **80.6%** on imbalanced data
• Explored **Dense and ResNet-style networks**, reaching **74.0% recall** and **77.2% ROC-AUC**, though CatBoost led overall

Flight Delay Prediction and Rescheduling Optimization | Open IIT - Data Analytics [Nov'24]
• Preprocessed a **3M-row** flight dataset, retaining **21%** of records with valid delay reasons for predictive modelling
• Constructed relevant time and operational features, removed outliers (**IQR**), and label-encoded categorical fields
• Configured **XGBoost** for delay prediction, achieving a **MAE of 9.5 mins**; explored a **simple LSTM** for comparison
• Designed a **Genetic Algorithm** to reschedule flights, reducing cumulative delay over **100** generations via evolutions

SKILLS AND EXPERTISE

Languages: C | C++ | Java | Python | HTML | CSS | MATLAB
Software: IBM ILOG CPLEX | SolidWorks | VS Code | MS Excel | Spyder | SketchUp | Pycharm
Library and Framework: NumPy | Pandas | Matplotlib | Gurobi | Folium | Ortools | Pymoo | Seaborn

COURSEWORK INFORMATION

Programming and Data Structures* | Probability and Statistics | Advanced Calculus | Linear Algebra & Numerical Analysis | Operations Research* | Basic Electronics* | Management of Inventory Systems | Economics | Work-System Design*
**(courses with theory and lab components)*

POSITIONS OF RESPONSIBILITY

Associate Member | ProDex [Jun'24-Jun'25]
• Ranked among **top 8** teams in **Empower Design Challenge**, for prototyping a device for wheelchair-to-car transfer
• Facilitated **5+** technical workshops and mentored juniors on project building, upskilling, and competitive readiness

EXTRA CURRICULAR ACTIVITIES

• Participated in **Kharagpur Data Science Hackathon**, applying ML and data analytics to real-world challenges
• Developed the frontend interface for AI chat bot in **GC OpenSoft** & pitched product concepts in **GC Product Design**
• Formally trained in **dance, painting**; represented school in **elocution**; participated in **Rangoli** at Spring Fest 2024